

# Venu Gopal Agarwal

Postdoctoral Researcher · Energy Sciences, EPFL

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## PROFILE

Computational researcher with a PhD in Energy Sciences from EPFL (2026), now a postdoctoral researcher at EPFL developing multiphysics models of electrochemical energy-conversion devices. Experienced in formulating and implementing coupled, nonlinear transport models (charge, species, heat and momentum), transient models for stability and degradation, and validating simulations against experiments. Strong scientific programming in Python, C/C++ (including a parallel MPI solver written from scratch) and Fortran.

## RESEARCH EXPERIENCE

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| Sept 2026 – present  | <p><b>Postdoctoral Researcher · LRESE, EPFL</b></p> <p>Multiphysics modelling towards efficiency and stability improvement of gas-fed CO<sub>2</sub> electrolysis (Prof. Sophia Haussener)</p> <ul style="list-style-type: none"><li>Modelling zero-gap membrane-electrode-assembly CO<sub>2</sub> electrolyzers with gas diffusion electrodes, resolving strongly coupled, nonlinear multiphysics across length scales, and developing transient models for stability diagnostics and improvement.</li></ul>                                                                                                                                                                                                                                                                                                                         |
| June 2021 – Aug 2026 | <p><b>Doctoral Assistant · LRESE, EPFL</b></p> <p>Quantitative assessment of transport limitations in CO<sub>2</sub> electrolyser systems (Prof. Sophia Haussener)</p> <ul style="list-style-type: none"><li>Developed a 2D gas-diffusion-electrode cathode model of a flow cell, resolving coupled charge, species and momentum transport; validated against experiments to identify performance-limiting mechanisms (Communications Chemistry, 2024).</li><li>Built a 1D bipolar-membrane model with electric-field-enhanced water dissociation; parametric and sensitivity studies to isolate rate-limiting transport and guide membrane design.</li><li>Coupled the GDE and bipolar-membrane models into an integrated bicarbonate-fed electrolyser model linking electrochemistry, ion transport and water management.</li></ul> |
| Jan – June 2020      | <p><b>Research Assistant · Thermofluids &amp; Energy Systems Lab, IIT Delhi</b></p> <ul style="list-style-type: none"><li>Data-driven, control-oriented reduced-order model (discrete-time realisation algorithm) to estimate the state of charge of a Li-ion cell in real time, accurate across standard drive cycles.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| May 2018 – Nov 2019  | <p><b>M.S. Research Scholar · Dept. of Mechanical Engineering, IIT Delhi</b></p> <ul style="list-style-type: none"><li>Wrote a parallel (MPI) lattice Boltzmann solver from scratch in C/C++; validated against in-house experiments (Physical Review Fluids, 2020).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## EDUCATION

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| 2021 – 2026 | <b>Ph.D. Energy Sciences</b> · École Polytechnique Fédérale de Lausanne (EPFL), Switzerland        |
| 2018 – 2019 | <b>M.S. (R) Thermal Engineering</b> · Indian Institute of Technology Delhi · GPA 10/10, Gold Medal |
| 2013 – 2017 | <b>B.Tech. Mechanical Engineering</b> · Indian Institute of Technology Gandhinagar · GPA 8.8/10    |

## PUBLICATIONS

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- V. G. Agarwal, S. Haussener, "Quantifying mass transport limitations in a microfluidic CO<sub>2</sub> electrolyzer with a gas diffusion cathode," *Communications Chemistry* 7, 47 (2024). doi:10.1038/s42004-024-01122-5
- M. Wada, K. Obata, V. G. Agarwal, F. Lorenzutti, S. Haussener, K. Takanabe, "Numerical analysis on field-enhanced homogeneous dissociation of protonated buffer species within the electrical double layer during hydrogen evolution reaction," *ACS Electrochemistry* 1, 2521 (2025). doi:10.1021/acselectrochem.5c00299
- V. G. Agarwal, R. Singh, S. S. Bahga, A. Gupta, "Dynamics of droplet formation and flow regime transition in a T-shaped microfluidic device with a shear-thinning continuous phase," *Physical Review Fluids* 5, 044203 (2020). doi:10.1103/PhysRevFluids.5.044203
- R. Saini, A. De, V. Agarwal, R. Yadav, "Investigation of the role of chemical kinetics in controlling stabilization mechanism of the turbulent lifted jet flame using multi-flamelet generated manifold approach," in *Energy for Propulsion*, Springer (2018). doi:10.1007/978-981-10-7473-8\_12

## AWARDS

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- EPFLglobalLeaders Doctoral Fellowship (Marie Skłodowska-Curie, Horizon 2020): competitive 48-month EU inter-sectoral fellowship, 2021.
- Gold Medal, perfect GPA (10/10), M.S. Thermal Engineering, IIT Delhi, 2020.
- 5 conference presentations, including an invited talk at ECS 2024 (Electrochemical Society); best poster award, IIT Kanpur, 2017.

## SKILLS

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| Modelling   | Coupled multiphysics PDEs, electrochemistry and porous-media transport, lattice Boltzmann, parameter calibration, sensitivity and scenario analysis |
| Programming | Python, C/C++ & MPI, Fortran, MATLAB                                                                                                                |
| Tools       | COMSOL Multiphysics, Ansys Fluent                                                                                                                   |